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Math 208 Final Exam Term 233

Original exam date: 15 August 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Using variation of parameters, the differential equation $y'' + 9y = \frac{1}{4}\csc(3x)$ has particular solution $y_p(x) = u_1(x)\cos(3x) + u_2(x)\sin(3x)$. Then $u_1(x) =$

(a) $x^2 - 1$

(b) $\frac{1}{3}x$

(c) $x + 1$

(d) $-\frac{1}{12}x$

(e) $4x$

2. If the matrix $A = \begin{bmatrix} 5 & -6 \\ 2 & -2 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 2 & 3 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $P = \begin{bmatrix} -2 & 3 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(e) $P = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

3. A possible fundamental matrix for the system $X' = \begin{bmatrix} 5 & -3 \\ 2 & 0 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} 3e^{2t} & 3e^{3t} \\ 4e^{2t} & 2e^{3t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} e^{2t} & 3e^{3t} \\ e^{2t} & 2e^{3t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} e^{2t} & 4e^{3t} \\ e^{2t} & 2e^{3t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} e^{2t} & e^{3t} \\ e^{2t} & 2e^{3t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} 2e^{2t} & 3e^{3t} \\ e^{2t} & 2e^{3t} \end{bmatrix}$

4. The solution of the exact differential equation

$$\left(\frac{y}{x} + y + \ln y\right) dx + \left(\frac{x}{y} + x + \ln x\right) dy = 0$$

is

(a) $y^2 \ln x + xy + x \ln y = c$

(b) $y \ln x + xy^2 - x \ln y = c$

(c) $y^2 \ln x + x^2 y + x \ln y = c$

(d) $y \ln x + xy + x \ln y = c$

(e) $y \ln x + x^2 y + x \ln y = c$

5. If the rank of the matrix $A = \begin{bmatrix} 1 & -3 & -9 & -5 \\ 2 & 1 & K & 11 \\ 1 & 3 & 3 & 13 \end{bmatrix}$ is 2, then $K =$

(a) 4

(b) -5

(c) 5

(d) -4

(e) 6

6. The sum of the indicial roots at $x = 0$ for the differential equation $3xy'' + y' - y = 0$ is

(a) $-\frac{3}{2}$

(b) $\frac{3}{2}$

(c) 0

(d) $\frac{2}{3}$

(e) $-\frac{2}{3}$

7. A general solution of the system $X' = \begin{bmatrix} 1 & -4 \\ 4 & 9 \end{bmatrix} X$ is

(a) $X = \left(c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 2+t \\ -t \end{bmatrix} \right) e^{5t}$

(b) $X = \left(c_1 \begin{bmatrix} -4 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 1-4t \\ 4t \end{bmatrix} \right) e^{5t}$

(c) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1-t \\ -t \end{bmatrix} \right) e^{5t}$

(d) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1+4t \\ -4t \end{bmatrix} \right) e^{5t}$

(e) $X = \left(c_1 \begin{bmatrix} 4 \\ -4 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -4t \end{bmatrix} \right) e^{5t}$

8. An appropriate form of particular solution of the differential equation

$$y'' - 2y' + 2y = e^x \sin x$$

is

(a) $y_p = Axe^x \cos x + Bxe^x \sin x$

(b) $y_p = Ax^2e^x \cos x + Bx^2e^x \sin x$

(c) $y_p = Axe^{-x} \cos x + Bxe^{-x} \sin x$

(d) $y_p = Ae^x \cos x + Be^x \sin x$

(e) $y_p = Ae^{-x} \cos x + Be^{-x} \sin x$

9. Let $F(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $e^{At} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$. A particular solution for the system $X' = AX + F(t)$ is

(a) $X_p = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

(b) $X_p = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$

(c) $X_p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $X_p = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$

(e) $X_p = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$

10. By making a suitable substitution, the differential equation $y' = (9x + 4y)^2$ can be transformed into a separable differential equation

(a) $v' = 9v^2 + 4$

(b) $v' = 9v^2 - 4$

(c) $v' = 9v^3 - 4$

(d) $v' = 4v^2 - 9$

(e) $v' = 4v^2 + 9$

11. If $y = \sum_{n=0}^{\infty} c_n x^n$ is a power series solution about the ordinary point $x = 0$ of the differential equation $y'' + xy = 0$, then the coefficients c_n satisfy

(a) $c_{n+2} = \dfrac{-2c_n}{(n+1)(n+2)},\ n \geq 1$

(b) $c_{n+2} = \dfrac{2c_n}{(n+1)(n+2)},\ n \geq 1$

(c) $c_{n+2} = \dfrac{-c_{n-1}}{(n+1)(n+2)},\ n \geq 1$

(d) $c_{n+2} = \dfrac{-c_n}{n(n+2)},\ n \geq 1$

(e) $c_{n+2} = \dfrac{-c_n}{(n+1)(n+2)},\ n \geq 1$

12. The matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$ has

(a) one eigenvalue of multiplicity 3

(b) no real eigenvalues

(c) Three distinct real eigenvalues

(d) one eigenvalue of multiplicity 1 and one eigenvalue of multiplicity 2

(e) one real and one pair of non real eigenvalue

13. If $A = \begin{bmatrix} 2 & 5 \\ 0 & 2 \end{bmatrix}$, then $e^{At} =$

(a) $\begin{bmatrix} e^{2t} & t^2e^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(b) $\begin{bmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{bmatrix}$

(c) $\begin{bmatrix} e^{2t} & 5e^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(d) $\begin{bmatrix} e^{2t} & 5 + te^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(e) $\begin{bmatrix} e^{2t} & 5te^{2t} \\ 0 & e^{2t} \end{bmatrix}$

14. A 2×2 real matrix A has an eigenvector $\begin{bmatrix} 3 \\ 1 - i \end{bmatrix}$ associated with the eigenvalue $\lambda = 2 + 3i$ of A. Then the general solution of the system $X' = AX$ is

(a) $X = c_1 \begin{bmatrix} 3\cos(3t) \\ 2\cos(3t) - \sin(3t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2\sin(3t) \\ -\cos(3t) + \sin(3t) \end{bmatrix} e^{2t}$

(b) $X = c_1 \begin{bmatrix} 3\cos(3t) \\ 2\cos(3t) - \sin(3t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} \sin(3t) \\ -\cos(3t) + \sin(3t) \end{bmatrix} e^{2t}$

(c) $X = c_1 \begin{bmatrix} 3\cos(3t) \\ \cos(3t) + \sin(3t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 3\sin(3t) \\ -\cos(3t) + \sin(3t) \end{bmatrix} e^{2t}$

(d) $X = c_1 \begin{bmatrix} 3\cos(3t) \\ \cos(3t) - \sin(3t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} \sin(3t) \\ -\cos(3t) + \sin(3t) \end{bmatrix} e^{2t}$

(e) $X = c_1 \begin{bmatrix} \cos(3t) \\ \cos(3t) + \sin(3t) \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 3\sin(3t) \\ -\cos(3t) + \sin(3t) \end{bmatrix} e^{2t}$

15. The general solution of the differential equation

$$(D-1)(D+2)^2(D^2+9)y = 0$$

is given by

(a) $y = c_1e^x + c_2e^{-2x} + c_3xe^{-2x} + c_4\cos(3x) + c_5\sin(3x)$

(b) $y = c_1e^x + c_2e^{-2x} + c_3\cos(3x) + c_4\sin(3x)$

(c) $y = c_1e^x + c_2e^{2x} + c_3xe^{2x} + c_4\cos(3x) + c_5\sin(3x)$

(d) $y = c_1e^{-x} + c_2e^{2x} + c_3xe^{2x} + c_4\cos(3x) + c_5\sin(3x)$

(e) $y = c_1e^x + c_2xe^{-2x} + c_3\cos(3x) + c_4\sin(3x)$

16. Let $u = (7, -6, 4, 5)$, $v = (3, -3, 2, 3)$ and $w = (1, 0, 0, -1)$ be three vectors in $\mathbb{R}^4$. If $w = c_1u + c_2v$, then $c_1 + c_2 =$

(a) 3

(b) 0

(c) -1

(d) 1

(e) -2

17. If $y(x)$ is the solution of the initial value problem

$$\frac{dy}{dx} = 6e^{2x-y}, \qquad y(0) = 0,$$

then $y(\ln 2) =$

(a) $\ln 8$

(b) $\ln 10$

(c) $\ln 6$

(d) $\ln 12$

(e) $\ln 4$

18. The solution of the linear differential equation $xy' = 2y + x^3e^{-x}$ is

(a) $y(x) = -x^2e^{-x} + cx$

(b) $y(x) = -x^2e^{-x} + cx^2$

(c) $y(x) = x^2e^x + cx^2$

(d) $y(x) = -x^2e^{-x} + cx^3$

(e) $y(x) = x^2e^x + cx$

19. The general solution of the system $X' = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} X$ is given by

(a) $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t}$

(b) $X = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t}$

(c) $X = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t}$

(d) $X = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{5t}$

(e) $X = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{5t}$

20. The guaranteed radius of convergence of the power series solution of the differential equation $y'' + 4xy' + 2y = 0$ about the ordinary point $x = 0$ is

(a) 2

(b) 0

(c) 3

(d) ∞

(e) 1

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$W = 3$, $u_1' = -y_2 f/W = -\frac{1}{12}$, so $u_1 = -\frac{x}{12}$.
2	**(d)**	$\lambda = 2, 1$ with eigenvectors $(2,1), (3,2)$: $D = \mathrm{diag}(2,1)$.
3	**(b)**	$\lambda = 2, 3$ with eigenvectors $(1,1), (3,2)$ as columns times $e^{\lambda t}$.
4	**(d)**	$F = \int M\,dx = y\ln x + xy + x\ln y$, and $F_y = N$ so $g' = 0$.
5	**(d)**	$R_2 - 2R_1 \parallel R_3 - R_1 = 6(0,1,2,3)$ forces $K + 18 = 14$, so $K = -4$.
6	**(d)**	$3r(r-1) + r = 3r^2 - 2r = 0$: roots $0, \frac{2}{3}$, sum $\frac{2}{3}$.
7	**(b)**	$\lambda = 5$ (double), $\mathbf{v} = (-4, 4)$, $\mathbf{w} = (1, 0)$: second vector $(1 - 4t,\ 4t)$.
8	**(a)**	$r = 1 \pm i$ matches the forcing, so multiply the trial by x.
9	**(a)**	$X_p = e^{At}\int e^{-At}F\,dt = e^{At}(-\cos t, \sin t)^T = (-1, 0)^T$.
10	**(e)**	$v = 9x + 4y$: $v' = 9 + 4y' = 9 + 4v^2$.
11	**(c)**	Shifting indices: $(n+2)(n+1)c_{n+2} + c_{n-1} = 0$ for $n \geq 1$.
12	**(c)**	$\det(A - \lambda I) = -\lambda(\lambda - 3)(\lambda + 4)$: three distinct real eigenvalues.
13	**(e)**	$A = 2I + N$, $N^2 = 0$: $e^{At} = e^{2t}(I + Nt)$.
14	**(c)**	$\mathbf{a} = (3,1)$, $\mathbf{b} = (0,-1)$: $X_1 = e^{2t}(\mathbf{a}\cos 3t - \mathbf{b}\sin 3t)$, $X_2 = e^{2t}(\mathbf{b}\cos 3t + \mathbf{a}\sin 3t)$.
15	**(a)**	Roots $1, -2, -2, \pm 3i$: $e^x,\ e^{-2x},\ xe^{-2x},\ \cos 3x,\ \sin 3x$.
16	**(c)**	$c_2 = -2c_1$, $7c_1 + 3c_2 = 1$ gives $c_1 = 1$, $c_2 = -2$; sum -1.
17	**(b)**	$e^y\,dy = 6e^{2x}dx$: $e^y = 3e^{2x} - 2$, so $y(\ln 2) = \ln 10$.
18	**(b)**	$\rho = x^{-2}$: $(y/x^2)' = e^{-x}$, so $y = -x^2e^{-x} + cx^2$.
19	**(a)**	$\lambda = -1, 5$ with eigenvectors $(1,-1)$, $(1,2)$.
20	**(d)**	$P = 4x$, $Q = 2$ are polynomials, so the radius is ∞.